

What Should SRE Watch?

Calibration-Gated Decisions and the Missing Golden Signals for GPU Clusters

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scan for live demo

Queue wait is the user-visible latency on a shared GPU cluster. GPU utilization does not see it. We measure Queue-of-Work Reliability (QWR) — a calibrated $P(\text{wait} > \text{SLO})$ — and gate actions on its ECE. Live model ECE 0.068, held-out Slurm ECE 0.056: both land in T2 Warn.

THE FOUR TRAVEL POORLY

Classical signals

Google SRE’s four — latency, traffic, errors, saturation — assume a stateless request/response service. On shared GPU clusters:

Latency has token, batch, epoch, and queue meanings. Submit-to-start often dominates. HTTP p99 never sees it.

Traffic as QPS is undefined for training. Tokens/s and active jobs are different loads.

Errors as HTTP 5xx miss silent quality failure (200 + a bad output) and OOM / throttle restarts.

Saturation as nvidia-smi utilization is time-busy, not useful compute. Memory-bound kernels, thermal trips, and NCCL stragglers can all look fully utilized.

Mechanistic failure modes. The Slurm DCGM slice is idle T4 queue-wait (utilization = 0); a util-goodput correlation is undefined on that trace.

A PAGING-ROTATION MINIMUM

Five candidates

QWR latency / queue
At submit: $\hat{p} = P(\text{wait} > \text{SLO})$. Rolling ECE of $\hat{p}$.
T3 Suggest < 0.05; T4 Gate < 0.03.
Measured — held-out Slurm + live VGAC

GCE saturation / compute
Util dispersion, thermal spread, power-brake rate. Utilization is one feature, not the signal.
Defined — needs non-idle DCGM

HPH saturation / HBM
Headroom remaining plus its derivative, catching OOM cliffs before they page.
Defined — same idle-trace limit

SAI latency / collectives
 P_{90}/P_{50} step time across workers.
Candidate — no per-worker step times

OQC errors / quality
ECE of *model outputs* vs a shadow label stream. Not the queue-delay classifier.
Candidate — not measured here

THE TITLE CLAIM

Both models: T2 Warn

Actions are not “the model said 70%.” They are gated on whether that 70% is calibrated. The tier ladder:

T1 Annotate	T2 Warn	T3 Suggest	T4 Gate
ECE < 0.10	ECE $\uparrow$ 0.07 both models	ECE < 0.05	ECE < 0.03

Live VGAC v4.0-richfeatures-lr
ECE **0.068** AUROC 0.766 $n = 5,184$
T2 Warn (fails 0.05 Suggest and 0.03 Gate)

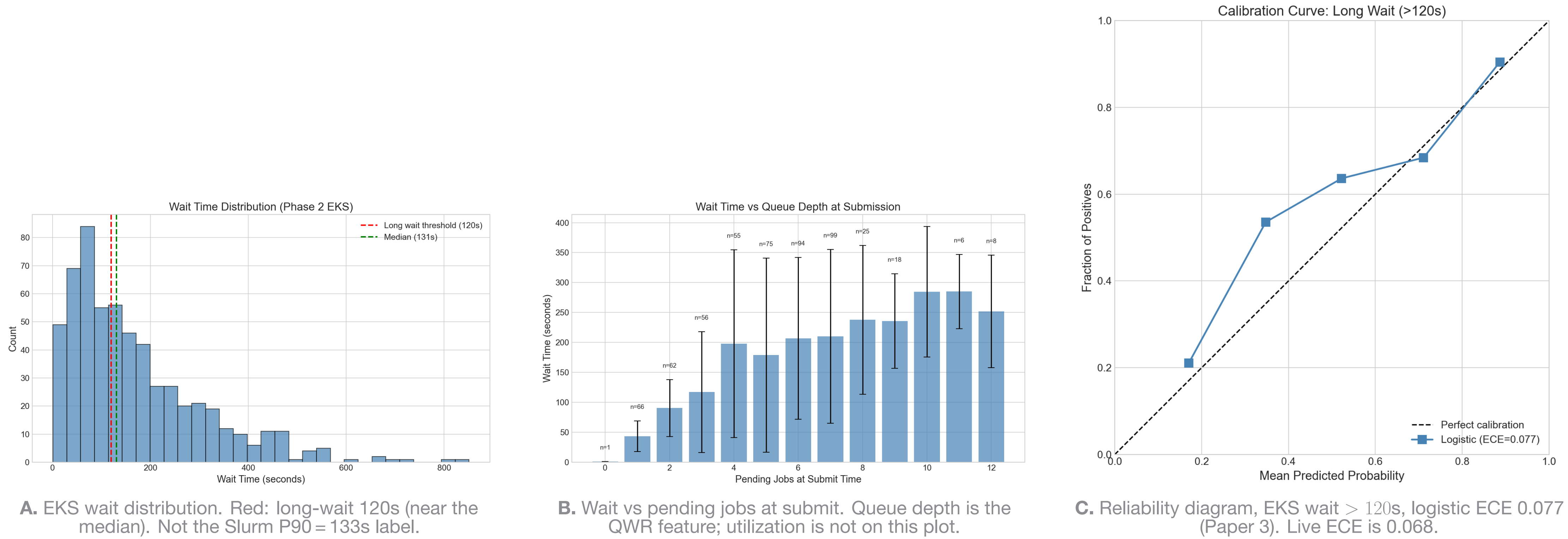
Held-out Slurm public artifact, 444/111 GB AUROC **0.863** (CI 0.71–0.96)
ECE **0.056** **T2 Warn**
~8 positives in test set 0 true alerts at $p \geq 0.7$

Labels differ between clusters

- EKS: wait > 120s (near the median).
 - Slurm: wait > P90 = 133s (~10% base rate).
- Do not pool AUROC or ECE across those labels.
Figure C is Paper 3 (650 EKS jobs, ECE 0.077) — not the live 0.068 card.

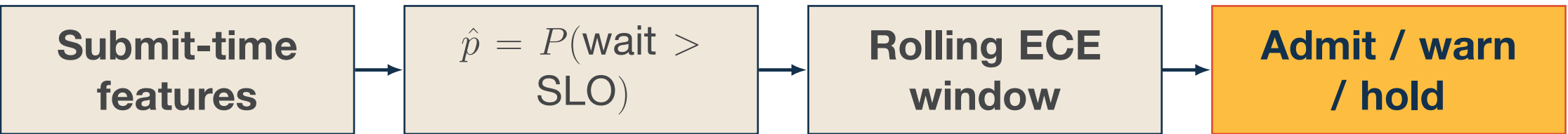
MEASURED EVIDENCE

Queue wait is the measured story



HOW AN SRE WOULD USE THIS

From $\hat{p}$ to a decision



VGAC at **demo.vgac.cloud** runs this loop on a t4g.small Graviton host. Predictions=QWR. Calibration=the gate (currently T2 Warn). GPUs=GCE/HPH candidates. HPC=SAI candidate.

SCOPE AND LIMITATIONS

Limits

- One signal measured (QWR). GCE, HPH, SAI, OQC are candidates only.
- Small test set: ~8 positives in 111 Slurm test jobs.
- DCGM columns idle on T4 queue-wait trace; GCE/HPH cannot be computed from it.
- EKS and Slurm use different “long wait” labels; results are not pooled.